

VEENA CHIDANAND BADIGER

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PROFILE SUMMARY

Electronics and Instrumentation Engineer with hands-on experience in low-power embedded and IoT systems, backed by a peer-reviewed journal publication on an AI-enabled Smart Electric Meter. Strong foundation in digital design, embedded systems, and edge intelligence, with a keen research interest in energy-efficient SoC architectures and FPGA-based edge AI acceleration.

RESEARCH INTERESTS

- 1. Low-Power SoC Architectures for IoT and Edge Devices:** Energy-efficient SoC design focusing on clock/power gating, low-leakage logic, and architecture-level optimization for next-generation smart IoT systems.
 - 2. Low-Power Memory & Compute Architectures for AI Edge Processing:** Design of optimized SRAM/memory subsystems and lightweight compute units to support quantized neural networks, enabling fast and power-efficient AI inference at the edge.
 - 3. FPGA-Based Edge AI Acceleration:** Development of real-time, low-power CNN and QNN hardware accelerators on FPGA platforms for tasks such as object detection and edge intelligence.
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EDUCATION

Bachelors in Electronics and Instrumentation

Dec.'22 - Jun.'26

JSS Academy of Technical Education, India (National Assessment and Accreditation Council grade:A+)

CGPA: 8.27

Relevant Coursework: VLSI Design, Digital Signal Processing, Digital Design and HDL, Embedded Systems, Basic Electronics, IOT, Computer Organization and Architecture, Operating Systems, Computer Communication Networks, Machine Learning, Control System, Analog Electronic Circuits.

PUBLICATIONS

SMART ELECTRIC METER SYSTEM. International Journal of Innovative Research In Technology (IJIRT),12(8), 2184-2188. Jan.'2026

This paper presents the design and implementation of a **Smart Electric Meter System** aimed at improving energy monitoring and management. The proposed system enables **real-time measurement of electricity consumption**, automated data transmission, and reduced manual intervention. By integrating embedded systems and communication technologies, the solution enhances billing accuracy, minimizes power theft, and supports efficient energy usage. The system demonstrates a cost-effective and scalable approach suitable for modern smart grid applications.

Link: <https://ijirt.org/article?manuscript=190146>

PROJECTS

Smart Electric Meter System

Aug. – Dec. '25

- Developed an AI-enabled smart energy meter using ESP32, IoT sensors (current/voltage), MQTT/serial communication, Python, and a web dashboard, enabling real-time power monitoring, remote access (Telegram Bot API), and reliable data logging.
- Implemented automated energy-management logic (presence-based lighting and temperature-based fan control), along with future monthly consumption prediction and real-time device-level load classification, delivering an efficient, low-power IoT energy-management system.

Sun Tracking Solar Panel

Feb. – Jun. '24

- Designed a dual-axis solar tracker using Arduino UNO, LDR sensors, servo motors, and real-time monitoring to dynamically adjust the panel toward maximum light intensity.
- Implemented a closed-loop tracking algorithm (Arduino C/C++), validating performance through real-time solar intensity and power measurements, achieving ~25% higher energy capture than fixed panels.

Fire Fighting Robot using Arduino UNO

Apr. – Jul. '23

- Developed a real-time fire-detection and autonomous extinguishing robot using Arduino UNO, IR flame sensors, L298N motor driver, DC motors, servo mechanism, and a water-pump module, enabling accurate flame localization and automated suppression.
 - Demonstrated reliable operation across varying flame intensities and distances, validating a robust fire-response prototype through consistent real-world performance tests.
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INTERSHIPS

PCB Design Intern

Jan.'26 – Apr.'26

SkillDzire Technologies Pvt. Ltd.

- Designed mixed-signal PCBs using KiCad, performing schematic capture and layout for servo drivers, LED circuits, sensor interfaces, and RF applications.
- Delivered manufacturable, signal-robust designs by applying signal integrity principles, optimized power routing, and DFM considerations.

Intern – FPGA & Embedded Systems

Bharat Electronics Limited

Feb.'26 – May'26

- Developed and tested an FPGA-based temperature-analysis system using TMP126 and FPGA processing modules.
- Worked on hardware–software interfacing, automation scripts, debugging, simulation and system validation.

VLSI Physical Design Trainee

Jul.'26 – Present

Maven Silicon

- Hands-on training in ASIC/VLSI design flow, Verilog HDL, CMOS, synthesis, STA, Linux, and Physical Design fundamentals.
- Practicing RTL design and synthesis using Design Compiler and Tcl.

Engineering Intern – System Validation

Sep'.26 – Present

Marelli

- Supporting testing and validation activities in an automotive electronics environment.
 - Gaining hands-on experience in debugging, documentation, and structured testing workflows.
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TECHNICAL SKILLS

Programming Languages: C, Python, Verilog

Simulation & Design Tools: Cadence Virtuoso, MATLAB, Keil µVision, Quartus Prime, ModelSim, Xilinx ISE, Tanner S-Edit

Operating Systems: Red Hat Linux, Windows

Embedded Systems & Instrumentation Tools: Arduino IDE, LabVIEW
